

# Betting Data Exploration

CSAS 2026

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```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.1      v stringr    1.5.2
v ggplot2    4.0.0      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(DataExplorer)
```

```
files <- list.files("Seasonal Data", full.names = TRUE)
full_data <- sapplly(files, read_csv)
```

Warning: One or more parsing issues, call `problems()` on your data frame for details,  
e.g.:

```
dat <- vroom(...)
problems(dat)
```

Rows: 380 Columns: 57

```
-- Column specification -----
```

Delimiter: ","

chr (8): Div, Date, HomeTeam, AwayTeam, FTR, HTR, Referee, B365AH

dbl (49): FTHG, FTAG, HTHG, HTAG, HS, AS, HST, AST, HF, AF, HC, AC, HY, AY, ...

```
i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
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i Use `spec()` to retrieve the full column specification for this data.
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```

Rows: 380 Columns: 68

-- Column specification -----

Delimiter: ","

chr (7): Div, Date, HomeTeam, AwayTeam, FTR, HTR, Referee

dbl (61): FTHG, FTAG, HTHG, HTAG, HS, AS, HST, AST, HF, AF, HC, AC, HY, AY, ...

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```

Rows: 380 Columns: 71

-- Column specification -----

Delimiter: ","

chr (7): Div, Date, HomeTeam, AwayTeam, FTR, HTR, Referee

dbl (64): FTHG, FTAG, HTHG, HTAG, HS, AS, HST, AST, HF, AF, HC, AC, HY, AY, ...

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i Specify the column types or set `show_col_types = FALSE` to quiet this message.
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-- Column specification -----

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Delimiter: ","
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dbl (64): FTHG, FTAG, HTHG, HTAG, HS, AS, HST, AST, HF, AF, HC, AC, HY, AY, ...

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i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 380 Columns: 74
-- Column specification -----
Delimiter: ","
chr (7): Div, Date, HomeTeam, AwayTeam, FTR, HTR, Referee
dbl (67): FTHG, FTAG, HTHG, HTAG, HS, AS, HST, AST, HF, AF, HC, AC, HY, AY, ...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 380 Columns: 68
-- Column specification -----
Delimiter: ","
chr (7): Div, Date, HomeTeam, AwayTeam, FTR, HTR, Referee
dbl (61): FTHG, FTAG, HTHG, HTAG, HS, AS, HST, AST, HF, AF, HC, AC, HY, AY, ...

```

```

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 381 Columns: 68
-- Column specification -----
Delimiter: ","
chr (7): Div, Date, HomeTeam, AwayTeam, FTR, HTR, Referee
dbl (61): FTHG, FTAG, HTHG, HTAG, HS, AS, HST, AST, HF, AF, HC, AC, HY, AY, ...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 380 Columns: 65
-- Column specification -----
Delimiter: ","
chr (7): Div, Date, HomeTeam, AwayTeam, FTR, HTR, Referee
dbl (58): FTHG, FTAG, HTHG, HTAG, HS, AS, HST, AST, HF, AF, HC, AC, HY, AY, ...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 380 Columns: 65
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Delimiter: ","
chr (7): Div, Date, HomeTeam, AwayTeam, FTR, HTR, Referee
dbl (58): FTHG, FTAG, HTHG, HTAG, HS, AS, HST, AST, HF, AF, HC, AC, HY, AY, ...

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Rows: 380 Columns: 65
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Delimiter: ","
chr (7): Div, Date, HomeTeam, AwayTeam, FTR, HTR, Referee
dbl (58): FTHG, FTAG, HTHG, HTAG, HS, AS, HST, AST, HF, AF, HC, AC, HY, AY, ...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 380 Columns: 62
-- Column specification -----
Delimiter: ","
chr (7): Div, Date, HomeTeam, AwayTeam, FTR, HTR, Referee
dbl (55): FTHG, FTAG, HTHG, HTAG, HS, AS, HST, AST, HF, AF, HC, AC, HY, AY, ...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.

```

```

Rows: 380 Columns: 106
-- Column specification -----
Delimiter: ","
chr   (7): Div, Date, HomeTeam, AwayTeam, FTR, HTR, Referee
dbl  (98): FTHG, FTAG, HTHG, HTAG, HS, AS, HST, AST, HF, AF, HC, AC, HY, AY,...
time  (1): Time

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 380 Columns: 106
-- Column specification -----
Delimiter: ","
chr   (7): Div, Date, HomeTeam, AwayTeam, FTR, HTR, Referee
dbl  (98): FTHG, FTAG, HTHG, HTAG, HS, AS, HST, AST, HF, AF, HC, AC, HY, AY,...
time  (1): Time

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 380 Columns: 106
-- Column specification -----
Delimiter: ","
chr   (7): Div, Date, HomeTeam, AwayTeam, FTR, HTR, Referee
dbl  (98): FTHG, FTAG, HTHG, HTAG, HS, AS, HST, AST, HF, AF, HC, AC, HY, AY,...
time  (1): Time

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 380 Columns: 106
-- Column specification -----
Delimiter: ","
chr   (7): Div, Date, HomeTeam, AwayTeam, FTR, HTR, Referee
dbl  (98): FTHG, FTAG, HTHG, HTAG, HS, AS, HST, AST, HF, AF, HC, AC, HY, AY,...
time  (1): Time

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 380 Columns: 106
-- Column specification -----
Delimiter: ","
chr   (7): Div, Date, HomeTeam, AwayTeam, FTR, HTR, Referee
dbl  (98): FTHG, FTAG, HTHG, HTAG, HS, AS, HST, AST, HF, AF, HC, AC, HY, AY,...
time  (1): Time

```

```

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 380 Columns: 120
-- Column specification -----
Delimiter: ","
chr      (7): Div, Date, HomeTeam, AwayTeam, FTR, HTR, Referee
dbl    (112): FTHG, FTAG, HTHG, HTAG, HS, AS, HST, AST, HF, AF, HC, AC, HY, AY...
time     (1): Time

i Use `spec()` to retrieve the full column specification for this data.
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```

```

s04_sub <- full_data$`Seasonal Data/Season02004.csv`
#DataExplorer::create_report(s04_sub)

```

## Season 2004 subset, Bet365 odds and NO Asian handicap, example cleaning and mutation

```

library(lubridate)
s04 <- s04_sub %>%
  select("Date", "HomeTeam", "AwayTeam", "FTHG", "FTAG", "FTR", "B365H", "B365D", "B365A") %>%
  mutate(Date = dmy(Date))

```

## Applying data cleaning to all datasets, then binding game dataset into one

```

games <- full_data %>%
  map(~ .x %>%
    select(Date, HomeTeam, AwayTeam, FTHG, FTAG, FTR, B365H, B365D, B365A) %>%
    mutate(Date = dmy(Date)) %>%
    mutate(Season = year(max(Date)))) %>%
  bind_rows()

```

## Team level dataset

```

home_team <- games %>%
  transmute(
    Date,
    Season,
    Team = HomeTeam,
    Opponent = AwayTeam,
    GoalsFor = FTHG,
    GoalsAgainst = FTAG,
    Home = 1,
    Odds = B365H,
    OddsDraw = B365D,
    Result = case_when(
      FTR == "H" ~ "W",
      FTR == "D" ~ "D",
      FTR == "A" ~ "L"
    )
  )

away_team <- games %>%
  transmute(
    Date,
    Season,
    Team = AwayTeam,
    Opponent = HomeTeam,
    GoalsFor = FTAG,
    GoalsAgainst = FTHG,
    Home = 0,
    Odds = B365A,
    OddsDraw = B365D,
    Result = case_when(
      FTR == "A" ~ "W",
      FTR == "D" ~ "D",
      FTR == "H" ~ "L"
    )
  )

team_level <- bind_rows(home_team, away_team)

```